

## 7.10 HYDRAULIC SYSTEM (see Fig. 7-15 and 7-16)

The hydraulic system, designed to reduce the main-rotor control forces, is constructed of modular components and mounted on the left side of the transmission compartment. It consists of two independent but mainly identical systems, of which System 1 is the primary operating system while System 2 is the stand-by reserve system. If necessary, operational control crosses over from the primary system to the stand-by system automatically and without delay. Both systems offer 100 % specific power support.

Pressure supply is provided by two hydraulic pumps which are permanently driven by the main transmission and which normally deliver full system pressure. The control of pressurization for the three actuators in each system is provided by two selector valves (one each system) which are linked by an interconnecting plate to ensure that only one system is active. The selector valves are mechanically coupled with and controlled by a hydraulic shuttle valve that works on the basis of pressure differentiation. Should the pressure of the primary system (System 1) fall below a certain level, the shuttle valve will position the selector valves to cause the stand-by system (System 2) to take over control (cross-over) and System 1 to become inactive.

The pressure of each hydraulic system is monitored by a pressure switch that, if system pressure falls below approx 70 %, will activate the respective **HYDR 1** or **HYDR 2** warning light.

### **EFFECTIVITY** *On hydraulic systems with Positive Cross-over Indication*

The **HYDR 1** warning light will also come on (together with the **HY BLOCK** warning light) whenever System 2 is activated.

### **EFFECTIVITY** *ALL*

A jammed servo valve in System 1 will cause an electric signal to be released by micro switches which, by means of a relay, trigger a solenoid valve. This solenoid valve causes an artificial loss of pressure by opening a reverse flow in System 1 so that System 2 is activated by the shuttle valve and System 1 is then deactivated (cross-over). The **HY BLOCK** warning light comes on whenever the relay controlling the solenoid valve is activated.

The micro switches, noted above, are designed to recognize a jammed servo valve and are only integrated in System 1. In System 2, however, a connection provided by springs will by-pass a jammed servo valve noticed by increased control forces about the affected axis.

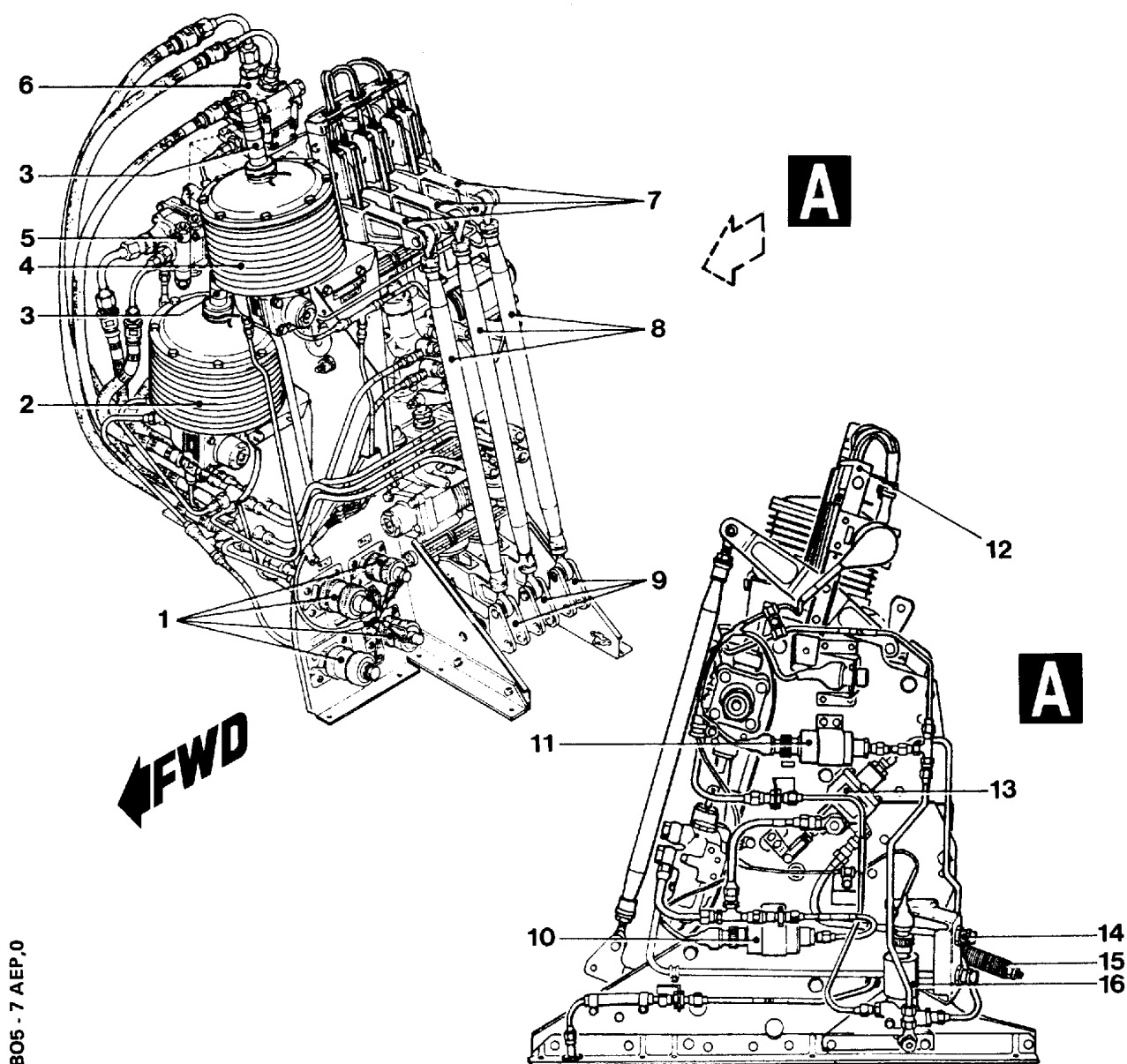
**NOTE** Should a servo valve jam in System 2, an automatic cross-over to System 1 is impossible. Therefore, normal operation in System 2 is not allowed as the then lack of redundancy would cause an inflight system failure to be a safety risk.

A cross-over functional check can be performed using a guarded three-position HYDR TEST switch located on the center console switch panel. Placing the switch from the

spring-loaded center (off) position to the TEST position activates the relay and, as a result, the solenoid valve to cause control to cross-over to System 2. Placing the switch to the RESET position deactivates the solenoid valve causing control to revert back to System 1.

All system control and monitoring functions are powered direct from the Main Bus via the HYDR. CONTROLS circuit breaker on the center console switch panel.





1. SERVICE COUPLINGS
2. RESERVOIR - SYSTEM 2
3. FLUID LEVEL INDICATORS
4. RESERVOIR - SYSTEM 1
5. HYDRAULIC PUMP - SYSTEM 2
6. HYDRAULIC PUMP - SYSTEM 1
7. INPUT LEVERS - SYSTEM 1
8. INTERCONNECTING CONTROL RODS

9. INPUT LEVERS - SYSTEM 2
10. PRESSURE SWITCH - SYSTEM 2
11. PRESSURE SWITCH - SYSTEM 1
12. MICRO SWITCHES
13. SHUTTLE VALVE
14. OPTIONAL EQUIPMENT COUPLINGS
15. DROOP COMPENSATION ASSEMBLY
16. SOLENOID VALVE

Fig. 7-15 Dual Hydraulic Module

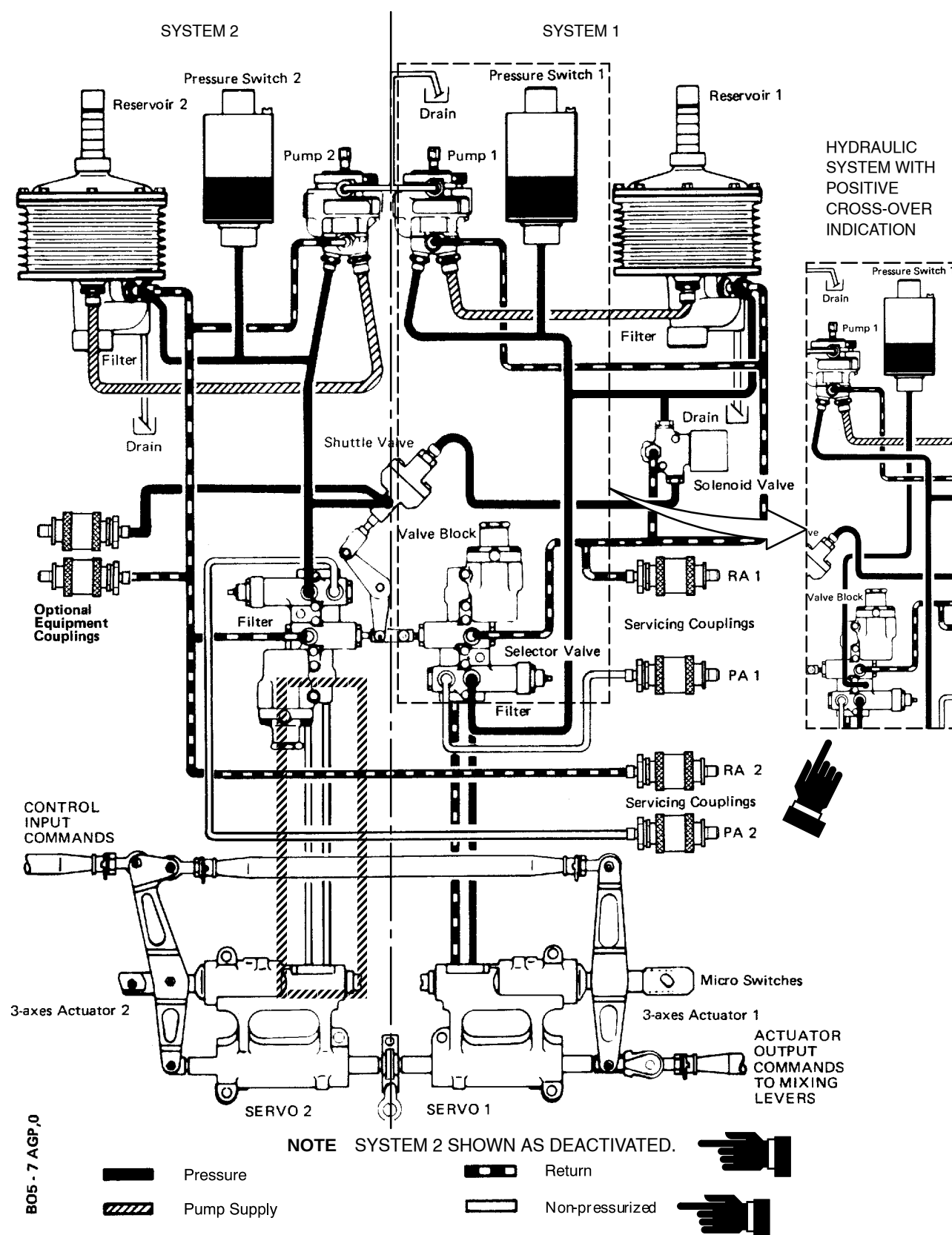


Fig. 7-16 Hydraulic System Schematic